

Phishing Email Detection & Awareness Report

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Introduction

Phishing is a social engineering attack where attackers send fraudulent emails that appear to come from legitimate organizations. These emails are designed to trick users into clicking malicious links, downloading infected attachments, or sharing sensitive information such as passwords, OTPs, or banking details.

Objective of the Task

The objective of this task is to analyze real-world style phishing emails using industry tools, identify phishing indicators, classify email risk levels, and create a security awareness document that helps users avoid phishing attacks.

Tools Used

The following tools and techniques were used during this analysis:

1. Public phishing email samples collected from security awareness blogs and datasets.
2. Google Email Header Analyzer to inspect email authentication results.
3. MXToolbox Email Header Analyzer to verify sender domain, SPF, DKIM, and IP reputation.
4. Browser hover inspection technique to safely analyze URLs without clicking them.

Each phishing email was analyzed by inspecting its sender address, email headers, message content, urgency level, and embedded links.

Phishing Email Sample 1

Phishing Email Example [GitHub] - [phishing_pot/email/sample-1022.eml](https://github.com/rf-peixoto/phishing_pot/blob/main/email/sample-1022.eml)
at main · [rf-peixoto/phishing_pot](https://github.com/rf-peixoto/phishing_pot)

Email Header Analysis (Using Google & MXToolbox)

MessageId	6e0aff7-e921-4f9b-a051-93760dc1e4ca@AM7EUR06FT021.eop-eur06.prod.protection.outlook.com
Created at:	8/2/2023, 6:17:50 PM GMT+5:30 (Delivered after 21 sec)
From:	shore@suksapan.or.th
To:	Recipients <shore@suksapan.or.th>
Subject:	Hi
SPF:	softfail with IP Unknown! Learn more
DKIM:	none Learn more
DMARC:	none Learn more

#	Delay	From *	To *	Protocol	Time received
0	16 sec	[102.69.139.111]	→ host4.ns.co.th		8/2/2023, 6:18:06 PM GMT+5:30
1	3 sec	AM7EUR06FT021.eop-eur06.prod.protection.outlook.com	→ AM6P191CA0081.outlook.office365.com		8/2/2023, 6:18:09 PM GMT+5:30
2	2 sec	AS8P189MB1509.EURP189.PROD.OUTLOOK.COM	→ AM6P191CA0062.outlook.office365.com		8/2/2023, 6:18:11 PM GMT+5:30

Phishing Indicators Identified

SPF failed

DKIM missing

DMARC missing

Unknown sender IP

Generic subject (“Hi”)

No sender verification

Risk Level: High Risk (Phishing Email)

Phishing Email Sample 2

Phishing Email Example [GitHub]- [phishing_pot/email/sample-1021.eml](https://github.com/rf-peixoto/phishing_pot/blob/main/email/sample-1021.eml) at main · rf-peixoto/phishing_pot

Email Header Analysis (Using MXToolbox)

MessageId	CANXpgOo_98j4Gfuj2mMFYUaTgWQhW4Gxu08VONmuQ9MZxc.JSDg@mail.gmail.com
Created at:	7/29/2023, 12:55:06 AM GMT+5:30 (Delivered after 78 sec)
From:	"Stephanie C. Frisble" <samranefahm@gmail.com>
To:	kjhgfdxgchvjbkkgfdxcvb@monkey56.skin
Subject:	_Get_Rid_of_Back_Pain_Easily_Try_Therapall_
SPF:	none Learn more
DKIM:	pass with domain Unknown! Learn more
DMARC:	fail Learn more

#	Delay	From *	To *	Protocol	Time received
0	15 sec		→ 2002:a05:6870:469f:b0:1ba:3ff0:493c		7/29/2023, 12:55:21 AM GMT+5:30
1		mail-sor-f65.google.com.	→ [Google] mx.google.com		7/29/2023, 12:55:21 AM GMT+5:30 <i>Originated at Gmail</i>
2			→ [Google] 2002:a9d:7445:0:b0:6b9:ed64:1423	SMTP	7/29/2023, 12:55:21 AM GMT+5:30
3	2 sec		→ [Google] 2002:a05:6808:14c8:b0:3a4:244d:fe3e	SMTP	7/29/2023, 12:55:23 AM GMT+5:30
4	11 sec		→ [Google] 2002:a05:6808:130e:b0:3a3:c78e:d863	SMTP	7/29/2023, 12:55:34 AM GMT+5:30
5			→ [Google] 2002:a4a:4146:0:b0:566:468c:c3b3	SMTP	7/29/2023, 12:55:34 AM GMT+5:30
6			→ [Google] 2002:a4a:8201:0:b0:566:f8ee:fa67	SMTP	7/29/2023, 12:55:34 AM GMT+5:30
7	4 sec		→ [Google] 2002:a05:6808:309d:b0:3a1:e818:f37b	SMTP	7/29/2023, 12:55:38 AM GMT+5:30
8	10 sec		→ [Google] 2002:a05:6808:219d:b0:3a1:f368:6b1	SMTP	7/29/2023, 12:55:48 AM GMT+5:30
9			→ [Google] 2002:a4a:4945:0:b0:56c:711b:1823	SMTP	7/29/2023, 12:55:48 AM GMT+5:30
10	35 sec		→ [Google] 2002:a05:6870:1f06:b0:1bb:81a1:f457	SMTP	7/29/2023, 12:56:23 AM GMT+5:30
11	1 sec		→ [Google] mail-oil-f208.google.com	SMTP	7/29/2023, 12:56:24 AM GMT+5:30
12		AM7EUR03FT054.eop-EUR03.prod.protection.outlook.com	→ AS9PR05CA0055.outlook.office365.com		7/29/2023, 12:56:24 AM GMT+5:30

Phishing Indicators Identified

SPF: none

→ No sender authorization found

DKIM: pass (but domain unknown)

→ Signature exists, but sender domain is not trusted

→ Domain policy check failed (strong phishing indicator)

Sender Name vs Email Mismatch

→ Display name looks real, but email is a random Gmail ID

→ Common impersonation technique

Suspicious Recipient Domain

→ Random and unusual domain (monkey56.skin)

→ Not linked to any legitimate service

Subject Line Pattern

→ Promotional, click-bait style

→ Uses underscores and marketing language

→ Common in spam/phishing emails

Email Routing Analysis

→ Originated from Gmail infrastructure

→ Gmail is often misused for mass phishing campaigns

Risk Level: Medium–High Risk (Phishing / Spam Email)

Security Awareness & Prevention Guidelines

Security Tips (What users SHOULD do)

1. Always check the sender's email address carefully.
Attackers often use fake or slightly changed domains.
2. Read the email slowly before taking any action.
Phishing emails try to create fear or urgency.
3. Hover over links before clicking on them.
If the link looks strange or unrelated, do not click.
4. Use official websites or apps to verify messages.
Never trust login links provided inside emails.
5. Report suspicious emails to the IT or security team.
Reporting helps protect other users as well.
6. Enable two-factor authentication (2FA) on important accounts.
This adds an extra layer of security.

What Users Should NOT Do

1. Do not click links from unknown or unexpected emails.
Even if the email looks urgent.
2. Do not share passwords, OTPs, or bank details through email.
Legitimate companies never ask for this.
3. Do not download attachments from unknown senders.
Attachments may contain malware.
4. Do not panic when you see messages like
"Account will be locked" or *"Immediate action required"*

7. Conclusion

This analysis demonstrates how phishing emails impersonate trusted brands and exploit urgency and fear. By using email header analysis tools and user awareness techniques, phishing attacks can be identified and prevented effectively.